

Forward & Backward Propagation

Worked example on an Artificial Neural Network — the XOR problem

Example Dataset (XOR Problem)

x1	x2	y (Label)
0	0	0
0	1	1
1	0	1
1	1	0

We'll use one training example: (**x1 = 0, x2 = 1, y = 1**).

Forward Propagation

1. Inputs

$$x_1 = 0, \quad x_2 = 1$$

2. Weights & Bias (initial guess)

$$w_1 = 0.5, \quad w_2 = 0.7, \quad b = -0.2$$

3. Weighted Sum

$$z = w_1 \cdot x_1 + w_2 \cdot x_2 + b$$

$$z = (0.5)(0) + (0.7)(1) + (-0.2) = 0.5$$

4. Activation (Sigmoid)

$$a = 1 / (1 + e^{-z}) = 1 / (1 + e^{-0.5}) \approx 0.62$$

5. Prediction

Output = 0.62 → interpreted as class **1**.

Backward Propagation

Backward propagation adjusts weights to reduce error.

1. Error (Loss)

True label $y = 1$, prediction $a = 0.62$.

$$L = \frac{1}{2}(y - a)^2 = \frac{1}{2}(1 - 0.62)^2 \approx 0.072$$

2. Gradient of Loss w.r.t. Output

$$\partial L / \partial a = (a - y) = (0.62 - 1) = -0.38$$

3. Gradient of Output w.r.t. Weighted Sum (Sigmoid derivative)

$$\partial a / \partial z = a(1 - a) = 0.62(1 - 0.62) \approx 0.235$$

4. Gradient of Loss w.r.t. z

$$\partial L / \partial z = (-0.38)(0.235) \approx -0.089$$

5. Weight Updates (Gradient Descent, $\eta = 0.1$)

$$w_1 = w_1 - \eta \cdot (\partial L / \partial z) \cdot x_1 \rightarrow \text{since } x_1 = 0, w_1 \text{ stays the same.}$$

$$w_2 = 0.7 - 0.1 \times (-0.089) \times 1 \approx 0.709$$

$$b = -0.2 - 0.1 \times (-0.089) \approx -0.191$$

Summary

- **Forward Propagation:** Compute prediction by passing inputs through weights + activation.
- **Backward Propagation:** Compute error, calculate gradients, update weights & bias to reduce error.
- After many iterations, the network learns the correct mapping.

Takeaway

- **Forward** = prediction phase
- **Backward** = learning phase (error correction)
- Together, they form the **training loop** of neural networks.

Diagram

Forward Propagation:

Inputs (x1, x2) --> Weights + Bias --> Activation --> Output (Prediction)

Backward Propagation:

Output Error --> Gradient Calculation --> Weight & Bias Updates --> Improved Prediction